

Check In

Input file:	standard input
Output file:	standard output
Time limit:	1 second
Memory limit:	1024 megabytes

In an ICPC contest, every participating team must first check in at the registration desk. You are asked to maintain the information system used for the sign-in process.

There is a roster containing the names of n teams. All team names are pairwise distinct, and every name consists only of uppercase or lowercase English letters.

After the roster, there are m queries. Each query gives a name (which also consists only of uppercase or lowercase English letters). For each query, you must classify it into exactly one of the following three outcomes:

- ****Successful sign-in**** — the name belongs to a valid team, and that team has not signed in before.
- ****Not a valid team**** — the name does not belong to any team in the roster.
- ****Already signed in**** — the name belongs to a valid team, but that team has already signed in earlier.

Input

The first line contains two integers n and m ($1 \leq n, m \leq 10^5$) — the number of teams in the roster and the number of queries.

The next n lines each contain a team name. All team names are distinct.

The next m lines each contain a query name.

Every name (both roster names and query names) consists only of uppercase or lowercase English letters. The total number of characters over all names (roster names and query names combined) does not exceed 10^6 .

Output

For each of the m queries, print one line containing exactly one of the following words:

- **'OK'** — if the query results in a successful sign-in;
- **'WRONG'** — if the name is not a valid team;
- **'REPEAT'** — if the name has already signed in.

Example

standard input	standard output
5 9	OK
RebornAsaVegetableDog	OK
LetItRot	OK
NaiveBirds	OK
OpenYourBrain	OK
CaptainTakesMeNot	REPEAT
CaptainTakesMeNot	WRONG
OpenYourBrain	WRONG
NaiveBirds	REPEAT
LetItRot	
RebornAsaVegetableDog	
NaiveBirds	
OneStepAway	
Metempsychosis	
CaptainTakesMeNot	